Enhancing smaller-scale magnetic structures in EMAG2 data

New York and adjacent US Atlantic margin | Processing workflow and preliminary interpretation

1. Rationale

Magnetic anomalies along the eastern US seaboard are influenced by large regional sources, including the East Coast Magnetic Anomaly (ECMA), which may obscure the response of smaller igneous bodies, basement contacts and buried rift-basin structures. Previous modelling of the New York Bight, Long Island and Central Bight basins has shown that the contribution of individual magnetic sources is difficult to distinguish where strong adjacent anomalies dominate the observed field (Turney et al., 2026a, 2026b). Here, we apply a sequence of transformations to the EMAG2 compilation to examine whether shorter-wavelength features can be resolved and mapped more clearly. Suppressing longer-wavelength anomalies may also reveal local magnetic signatures associated with potential mineral deposits that are otherwise masked by regional sources.

2. Data preparation and interpolation

EMAG2 magnetic anomaly values were imported from an XYZ dataset and projected into WGS84 / UTM Zone 18N (EPSG:32618). The original sampling interval was approximately 2.8–3.0 km. A regular 1.5 km grid was produced using linear interpolation, with cells outside the data convex hull masked. Linear interpolation was preferred to cubic interpolation to minimise the introduction of artificial extrema. The finer output grid improves visualisation and numerical processing, but does not increase the intrinsic resolution of EMAG2.

3. Reduction to the pole

Reduction to the pole (RTP) was undertaken to reduce the asymmetry and spatial displacement of magnetic anomalies associated with the present-day geomagnetic field. A 2025 reference field with inclination of approximately +65.0° and declination of −12.3° was adopted for the study area; the declination was adjusted for local UTM grid convergence. RTP was calculated in the Fourier domain using a regularised filter, reflective padding and edge tapering to reduce numerical instability and boundary artefacts. The transformation assumes an approximately uniform inducing field and predominantly induced magnetisation. It therefore provides a useful basis for structural comparison, rather than a unique representation of source geometry.

4. Regional–residual separation

To reduce the influence of broad regional anomalies, the RTP grid was smoothed using a Gaussian filter with a standard deviation of 10 km. The smoothed regional component was subtracted from the original RTP field to generate a residual anomaly map. This operation preferentially suppresses longer wavelengths and enhances relatively local magnetic variations. The 10 km parameter represents the standard deviation of the smoothing kernel, not a sharp wavelength cutoff; its geological significance must therefore be assessed alongside alternative smoothing scales.

5. Edge enhancement

Two derivative products were calculated from the residual RTP grid. The total horizontal derivative (THDR), defined as the magnitude of the east–west and north–south magnetic gradients, highlights abrupt lateral changes that may coincide with magnetised body margins or geological contacts. The tilt derivative (TDR) combines the first vertical derivative with THDR through an arctangent normalisation, allowing weaker gradients to be compared more readily with stronger anomalies. Zero contours of TDR were examined alongside THDR maxima to identify spatially coherent candidate boundaries. Derivatives can amplify short-wavelength noise and artefacts introduced during interpolation, filtering and RTP.

EMAG2v3 dataset reference (verified 2017 release)

Meyer, B., Saltus, R. & Chulliat, A. (2017). EMAG2v3: Earth Magnetic Anomaly Grid (2-arc-minute resolution), version 3. NOAA National Centers for Environmental Information. https://doi.org/10.7289/V5H70CVX

Meyer, B., Chulliat, A. & Saltus, R. (2017). Derivation and error analysis of the Earth Magnetic Anomaly Grid at 2 arc min resolution version 3 (EMAG2v3). Geochemistry, Geophysics, Geosystems, 18, 4522-4537. https://doi.org/10.1002/2017GC007280

Results and preliminary geological interpretation

Four-panel comparison: original RTP, 10 km residual RTP, THDR and TDR

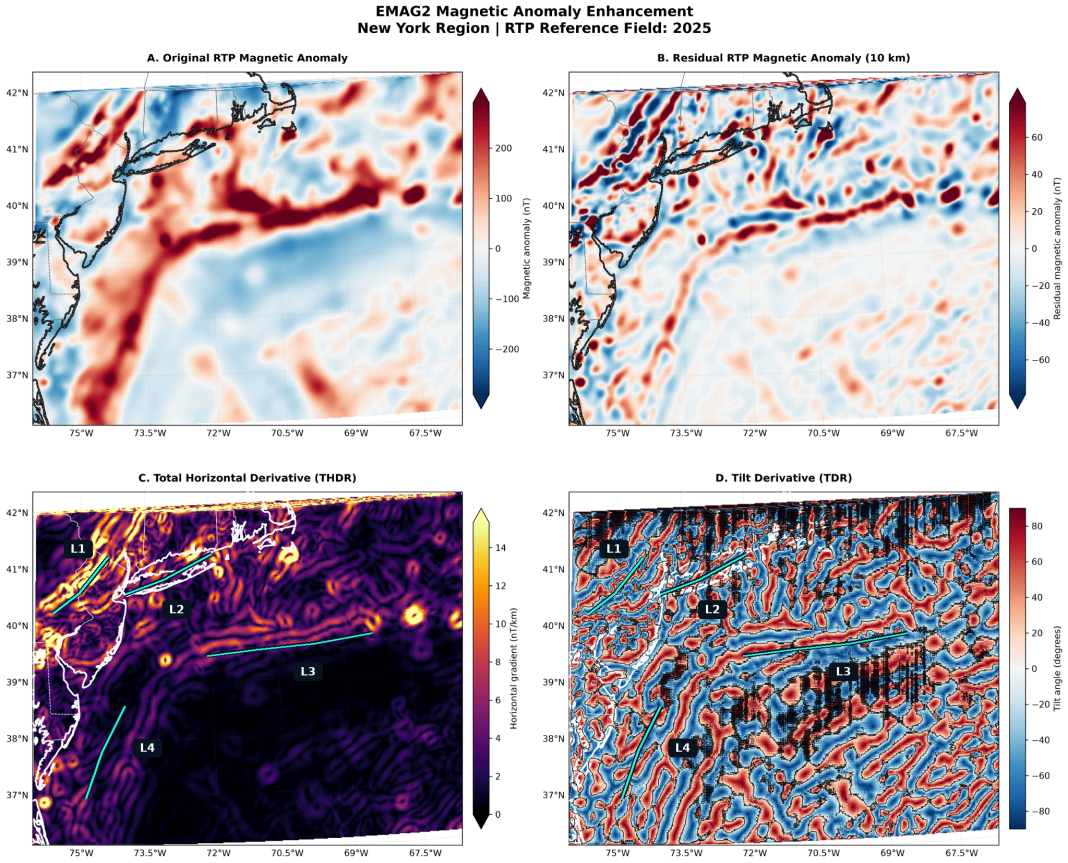


Figure 1. Original RTP, 10 km residual RTP, THDR and TDR maps with visually interpreted candidate magnetic lineaments L1-L4 (turquoise). Traces are approximate and have not been validated against mapped geology.

6. Interpretation

The original RTP field is dominated by broad positive magnetic anomalies, including an approximately east–west offshore corridor south of Long Island and a series of pronounced anomalies farther west. Regional–residual separation reveals additional, shorter and locally segmented anomalies that are less apparent in the unfiltered field. Corresponding THDR highs and changes in TDR identify several candidate magnetic lineaments, particularly in the northwestern onshore–nearshore sector, the central shelf and around the margins of the offshore magnetic corridor.

Some of these gradients may reflect lithological contacts, basement heterogeneity, intrusive bodies or fault-controlled changes in magnetisation. Their origin cannot be established from the derivative maps alone. The apparent continuity of narrow TDR features, particularly near grid edges, must be treated cautiously because differentiation can enhance processing artefacts. Moreover, the ~3 km original sample spacing limits the interpretation of the smallest features, regardless of the 1.5 km plotting interval.

7. Implications and next steps

Residual and derivative processing highlights candidate magnetic lineaments that may reflect contacts, intrusions or basement structures. However, these interpretations are preliminary. Geological maps, mapped faults and lithological boundaries, together with higher-resolution magnetic, gravity and seismic data, are required to test the location, continuity and geological origin of the proposed features. The illustrated traces should not be treated as confirmed faults or mapped geological contacts.

Selected context from the supplied publications

Turney, J.N. et al. (2026a). Identifying basaltic flows and sills in buried Mesozoic rift basins along the eastern US seaboard using seismic interpretation and geopotential modelling. Geophysical Journal International, 245, ggag085.

Turney, J.N., Goldberg, D.S. & Tinto, K.J. (2026b). New constraints on the structure of offshore Mesozoic rift basins and distribution of the Central Atlantic Magmatic Province along the eastern U.S. coastline from high-resolution aeromagnetic and gravity data. Frontiers in Earth Science, 14, 1800394.

Turney, J.N., Goldberg, D.S. & Tinto, K.J. (2026). Geopotential modeling of high-resolution survey data over the Great Stone Dome and near-shore diapiric structures. International Journal of Greenhouse Gas Control, 154, 104691.